

SHADI HAMDAN

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EDUCATION

Koç University, Turkey

September 2022 - August 2025

MSc in Computer Engineering, Autonomous Vision Group

Advisor: Asst. Prof. Fatma Güney

- Efficient, real-time sequential decision making approaches for real world applications (Autonomous Driving)
- Enabling utilization of heavy, expensive foundation model (VLM) usage with real-time considerations

Koç University, Turkey

July 2022

Bachelor of Science, Computer Engineering

Major GPA: 4.11/4.00, Overall: 4.08/4.00

- **Valedictorian**: ranked **first** among all 2022 graduates across all departments
- Full-ride scholarship from the Al Ghurair Foundation for Education

Northwestern University

January 2021 - March 2021

Global Exchange Program

GPA: 4.00/4.00

- Chosen to receive the single annual nomination to go on exchange for an academic semester to Northwestern.
- Achieved High Honors for the Winter 2021 quarter

WORK EXPERIENCE

Hyperbee.ai

July 2020 - October 2025

NLP Project Lead

Palo Alto, California (Remote)

- Designed efficient, IO-aware, NLP models optimized for use on edge devices
- Developed state of the art techniques to improve performance in targeted domains in machine translation, outperforming available solutions like Google Translate by 20% in terms of BLEU score

OpenDriveLab at Shanghai AI Lab

July 2024 - March 2025

Doctoral Research Intern (Principal Investigator: Asst. Prof. Hongyang Li)

Shanghai, China

- Dual-system approaches to autonomous driving with large (vision) language models
- Proposed an architecture that can utilize slow, heavy planner in tandem with a light-weight planner for SoTA performance in driving benchmarks with 6x speedup compared to previous approaches (**ICCV 2025**)

Koc University High Performance Computing Cluster

January 2023 - September 2023

Cluster Management Assistant

Istanbul, Turkey

- Prepared and gave PyTorch tutorials for multi-GPU training and mixed precision to over 80 graduate students
- Created bash and python scripts to help users track their cluster usage in order to efficiently utilize resources

Koc University and Is Bank Artificial Intelligence Center (KUIS-AI Center)

June 2020 - August 2022

Undergraduate AI Researcher

Istanbul, Turkey

- Worked on CorInfoMax, a self-supervised learning approach to learning visual representations based on information maximization. Accepted to **NeurIPS 2022**
- Curated MinuteMysteriesQA, a dataset of short "Whodunnit?" crime stories to evaluate the capability of models to perform complex reasoning. The dataset is now part of BIG-bench, which has been published in **TMLR**

PUBLICATIONS

- **S. Hamdan** et al., "ETA: Efficiency through Thinking Ahead, A Dual Approach to Self-Driving with Large Models", **ICCV 2025**
- D. Yuret, **S. Hamdan**, "What do LLMs learn from near-miss negative examples?," **on arXiv**
- **S. Hamdan**, F. Güney, "CarFormer: Self-Driving with Learned Object-Centric Representations," **in ECCV, 2024**.
- S. Ozsoy, **S. Hamdan**, Sercan Ö. Arik, D. Yuret, and Alper T. Erdogan, Self-Supervised Learning with an Information Maximization Criterion," **in NeurIPS, 2022**.
- A. Srivastava et al., "Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models," **on TMLR**.

PROJECTS

Skitt

Python Library

- Shadi's Toolkit (Skitt) is an open source PyTorch based library for optimizing model training and data loading
- Provides dataset caching, GPU and RAM usage profiling, and multi-GPU debugging and training utilities
- Has been used to optimize dataset loading and preprocessing on the fly by 2-5x on clusters with slow disk access

CorInfoMax

Research Project (NeurIPS 2022)

- Implemented the CorInfoMax self-supervised approach in PyTorch
- Implemented multi-GPU support for the CorInfoMax loss function
- Identified and addressed bottlenecks in dataloading as well as other improvements in the model that led to a two-fold speedup with minimal to no impact in performance

ACHIEVEMENTS

Valedictorian of the Class of 2022

June 2022

Awardee

Koç University, Istanbul, Turkey

- Ranked first among ~1300 graduates of 2022 from all departments.

Al Ghurair Foundation for Education STEM scholarship Program (AGFE)

September 2017 - Present

Recipient

Istanbul, Turkey

- A merit-based award for highly achieving Arab students.
- Admitted among ~400 students out of 20000+ applicants to pursue a fully funded STEM degree.

EXTRACURRICULAR ACTIVITIES

International Blockchain Olympiad 2020

July 2020

Participant

Online (due to Covid-19)

- Presented our Proof of Concept implementation of LightChain as part of DISNET Research Team

Koç University ACM Student Chapter (KUACM)

May 2019 - Present

Board Member

Istanbul, Turkey

- Organized programs and activities carried out under that chapter's supervision

Koç University Algorithms Program (KUAP)

January 2019 - Present

Instructor

Istanbul, Turkey

- Helped create a curriculum that covers basics of problem solving using algorithms and data structures
- Led discussions and problem solving sessions demonstrating concepts and solutions to problems

inzva Algorithmic Competition Winter Camp 2019

January 2019

Participant & Contestant

Istanbul, Turkey

- Ranked 1st in final day contest after a week long intensive training on algorithms and data structures

TEACHING EXPERIENCE

Programming Language Concepts

September 2022 – Present

Teaching assistant

Istanbul, Turkey

Deep Learning (Combined Graduate/Undergraduate Course)

September 2020 – January 2021

Teaching assistant

Istanbul, Turkey

- Conducted weekly lab sessions and office hours
- Graded assignments and term projects
- Assigned to guide 9 graduate and undergraduate students in their respective semester long project

Koç University Office of Learning and Teaching (KOLT)

September 2018 – June 2019

Teaching assistant

Istanbul, Turkey

- Conducted office hours, review sessions, and proctored exams for Linear Algebra and Advanced Programming.

SKILLS

- **Programming Languages:** Python (PyTorch, huggingface, deepspeed, scikit-learn, pandas), Julia (KNet), C++, C.

RELEVANT COURSEWORK

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|------------------------------|-------------------------------------|--------------------------------|
| • Deep Learning | • Linear System Theory | • Algorithm Design |
| • Computer Vision (CV) | • Natural Language Understanding | • Data Structures & Algorithms |
| • Intro. to Machine Learning | • Intro. to Artificial Intelligence | • Computer Architecture |
| • CV for Autonomous Driving | • Software Engineering | • Discrete Mathematics |